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From IGC records to measurable trajectories

Methods, results and questions for the supervisors

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Chapter 2 • Complete chapter review

This deck covers the whole chapter. It has no prescribed talk duration. The methodological choices, current results and unresolved validation questions are intended to be discussed together.

The archive contains recorded cross-country flights

	Paragliders	Hang gliders
Catalogue entries	203,343	9,259
Downloaded IGC logs	186,052	6,716
Retained flights	156,406	6,094
Retention	84.1%	90.7%

FFVL Coupe Fédérale de Distance: self-declared flights, heterogeneous recorders, sites and seasons.

Each decoded fix supplies

Time and horizontal position; GNSS and pressure altitude when recorded; receiver validity declaration.

Neither the catalogue nor the retained sample is an unbiased population of all soaring flights.

The analysis population is defined jointly by who records and declares a flight, and by which records our criteria retain.

Current pipeline census. Para catalogue seasons: 1999–2000 to 2025–2026; hang: 2001–2002 to 2025–2026. Downloaded-track coverage differs.

The archive contains recorded cross-country flights

The attempted and retained counts are imported from the current completed pipeline census. Catalogue entries, downloaded tracks and retained flights have different denominators. Participation, declarations, devices and the cross-country restrictions jointly select the observed population.

Acquisition preserves the declarations and their recorded tracks

Archived source

Season XML exports retain the original flight lists, metadata and track links. IGC files retain the recorder output.

Derived catalogue

One row per flight links its declared metadata to the local track and download status. Catalogue entries, linked tracks and readable downloads are different counts.

Resumable acquisition checks the recorder header and the presence of fix records. A readable file can still contain physically implausible measurements.

Which missing tracks and declaration choices could change comparisons across seasons or regions?

Acquisition preserves the declarations and their recorded tracks

The XML exports and IGC files supply the source evidence; the catalogue can be rebuilt. Acquisition checks file content, while subsequent cleaning assesses time, geometry and channel support. The analysis uses FFVL data only. Other platforms would change geographical and aircraft coverage but are not part of the current numerical population.

An IGC fix contains two altitude fields and a validity declaration

Record	Information used
A	Recorder identification
H	Date and descriptive headers
I	Optional extension-field layout
B	UTC time, latitude, longitude, GNSS validity, pressure altitude, GNSS altitude

A syntactically valid coordinate need not be a plausible measurement. The parser checks record structure; cleaning supplies the physical and temporal checks.

A V flag invalidates GNSS altitude here; horizontal position remains provisional until its own checks.

An IGC fix contains two altitude fields and a validity declaration

Explain the separation between syntax and observation quality. The B-record A/V declaration describes the GNSS solution, not the quality of every measured channel. A pressure value alone does not establish a usable barometer. Cadence is recorder-dependent and can vary within the original file.

Equipment classes and scoring labels describe different things

Equipment

Paraglider EN/LTF labels are harmonized; CCC, tandem/non-certified and unknown entries remain distinct. Hang-glider classes describe design and control.

Declared task

Free distance, out-and-return and triangle labels describe scoring categories. They do not label every manoeuvre or establish that a retained segment closes.

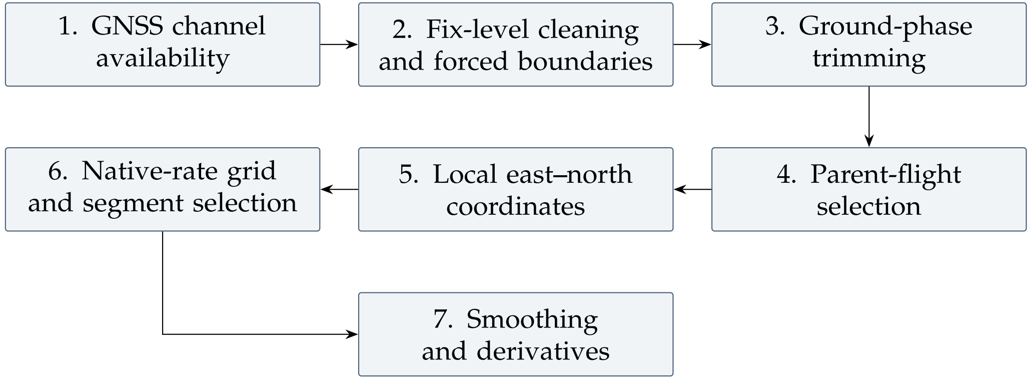
The later A/B versus C/D/CCC comparison uses equipment as an experience proxy. Actual pilot skill is not measured by that split.

What independent experience information would allow us to separate pilot choice from equipment?

Equipment classes and scoring labels describe different things

A class is neither a continuous performance score nor an independently observed skill level. An experienced pilot may use an A/B wing. Blank classes remain unknown rather than being assigned to beginners. Preserve the distinction when interpreting both regional anisotropy and duration composition in Chapter 3.

The processing order determines what each cut acts on



Parent-flight restrictions: recorded duration $\geq$ 40 min, path $\geq$ 20 km, altitude range $\geq$ 600 m, elapsed span $\leq$ 16 h, plus integrity and reach checks.

These restrictions select a study population. Genuine short flights or nearly level soaring can fail them. Flight-level cuts are not reapplied to every later segment.

The processing order determines what each cut acts on

Cleaning precedes smoothing. Trimming precedes the cross-country selection, so parked time is not counted as flown duration. Duration and path sum within contiguous blocks; elapsed span includes gaps. Fix-level rules can remove positions, invalidate only altitude or impose boundaries. Sampling later decides which remaining holes may be bridged. The 600 m altitude-range rule is a population restriction; it is not proof that an excluded flight was not soaring.

GNSS supplies altitude; pressure has a separate witness role

Adopted altitude channel

One channel throughout the population: GNSS presence $\geq$ 95% and range $\geq$ 30 m at flight level.

Missing or invalid GNSS altitude becomes missing z; horizontal coordinates remain subject to their own tests.

The barometer does not substitute for GNSS altitude.

Locally available corroboration

$$B_I = B_{\text{flight}} \prod_{i \in I} b_i.$$

Here $b_i = 1$ for a finite, nonzero pressure reading at fix i, and 0 otherwise.

The candidate interval needs simultaneous position and pressure readings. An incomplete local witness abstains.

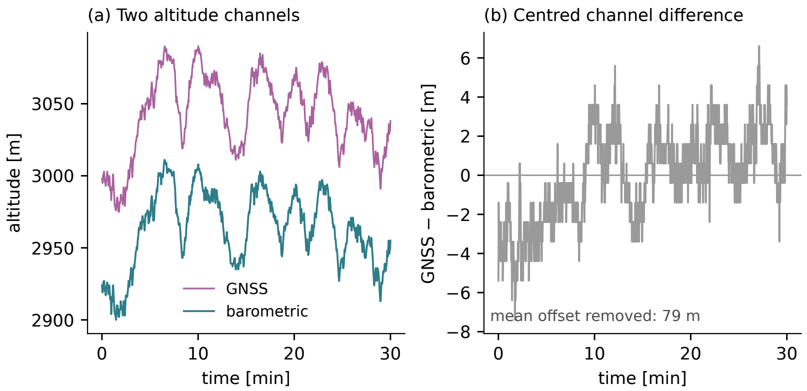
How sensitive are witness-dependent decisions to the availability of paired pressure observations?

Eligible barometer: flight presence $\geq$ 95%, range $\geq$ 30 m. Eligibility occurs in 72.1% para / 84.7% hang flights attempted by the pipeline.

GNSS supplies altitude; pressure has a separate witness role

The local pressure requirement answers the concern about GNSS lock. A pressure observation must accompany the horizontal fix when used; it must not be interpolated into the candidate interval. Valid GNSS altitude is not an additional requirement for that witness, since its failure may be the reason pressure is useful. A global rule discarding every unpaired fix would select devices and discard horizontal information unnecessarily. GNSS altitude is a consistent operational channel choice, not a claim that it is always less noisy than pressure altitude. Different height datums across recorders remain unresolved.

Recorded altitude traces combine flight motion and channel behaviour

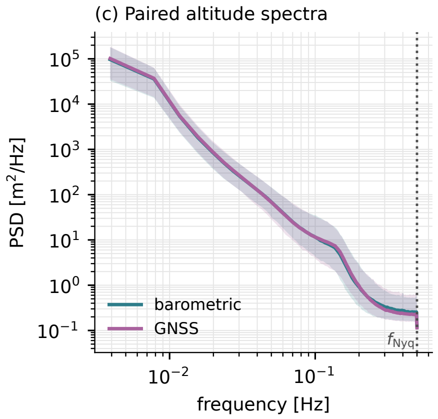


Which common variations plausibly reflect flight motion, and which differences need a sensor explanation?

Recorded altitude traces combine flight motion and channel behaviour

These are representative recorded traces, not ground-truth altitude. A difference between channels can involve datum, atmospheric effects, quantization or recording defects. The spectra and availability diagnostic use separately stated populations.

The paired PSD comparison uses raw valid intervals



Same valid block for both channels; linear resampling and Welch detrending.

No pipeline Savitzky–Golay smoothing enters these spectra.

Shading is the per-frequency 10–90% range across selected flights.

Similar median spectra do not establish identical sensors or identical variation between flights.

The paired PSD comparison uses raw valid intervals

The diagnostic begins with 3000 sampled raw paths per discipline and retains 2008 paired spectra under the stated support and cadence criteria. The longest paired regular block excludes missing-value codes, invalid GNSS flags and long gaps. The shaded per-frequency 10–90 percent range describes variation across these selected flights, not a confidence interval for sensor noise.

Similar typical PSD floors can conceal different upper tails

For each of the 2,008 paired flights, take the median PSD over 0.35–0.50 Hz. The table summarizes this quantity across flights.

Channel	Median	90th percentile	99th percentile
GNSS	2.24×10^{-1}	5.90×10^{-1}	1.38×10^{1}
Barometric	2.61×10^{-1}	5.08×10^{-1}	4.39

Spectral density in m^2/Hz.

The paired valid-block criteria condition which recorded intervals are measured. The PSD contains motion and recording effects; it is not a pure sensor-noise measurement.

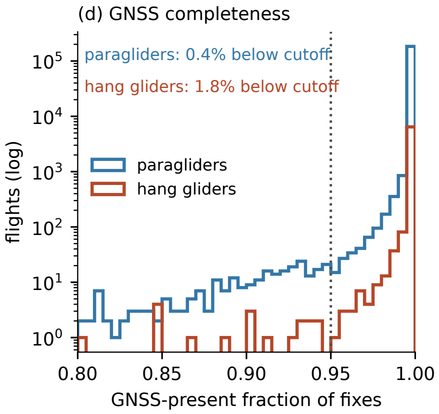
How sensitive are these tails to block length, cadence and logger type?

Current paired-block diagnostic. These floor percentiles differ from the per-frequency shaded range in the preceding plot.

Similar typical PSD floors can conceal different upper tails

No pipeline Savitzky–Golay filtering enters this calculation. Current median floors are about 0.224 GNSS and 0.261 barometric; the GNSS upper tail is larger, particularly at the 99th percentile. Similar visible bands therefore do not establish identical between-flight behaviour. Validity, paired availability and continuous support define the observation population. A narrower band cannot be interpreted as physical receiver improvement.

GNSS availability is a separate flight-level selection



The channel gate requires at least 95% usable GNSS altitude and at least 30 m range. The availability census is broader than the selected paired-spectrum population.

Which recorders, periods and flight types are preferentially lost through this requirement?

GNSS availability is a separate flight-level selection

The barometer remains an independent operational witness where it is eligible and locally complete. Its availability does not repair missing GNSS support in the cleaned vertical trajectory.

A common altitude channel does not resolve its vertical datum

The pipeline uses logged GNSS altitude throughout; it does not switch to pressure altitude where GNSS is absent.

$$z_{\text{logged}} = z_{\text{reference}} + b \quad \Rightarrow \quad \Delta z_{\text{logged}} = \Delta z_{\text{reference}} \quad \text{only when } b \text{ is constant.}$$

Recorder conventions and historical compliance remain unresolved. Spatially varying datum corrections and changing sensor biases do not cancel like a constant offset.

Which absolute-height or environmental comparisons need a separate datum audit?

A common altitude channel does not resolve its vertical datum

The consulted IGC-approved and CIVL documents describe different height conventions; the CIVL source contains draft annotations. The parser preserves the logged value, and actual archive compliance has not been established. Pressure altitude also depends on atmospheric pressure and temperature. Consistency of channel choice is not a guarantee that GNSS is uniformly more accurate.

Timestamp and coordinate bookkeeping precede geometric decisions

Duplicate timestamps are consolidated; backward clock steps, impossible coordinates and missing positions are treated explicitly. Exact position repetition also enters the frozen-run witness.

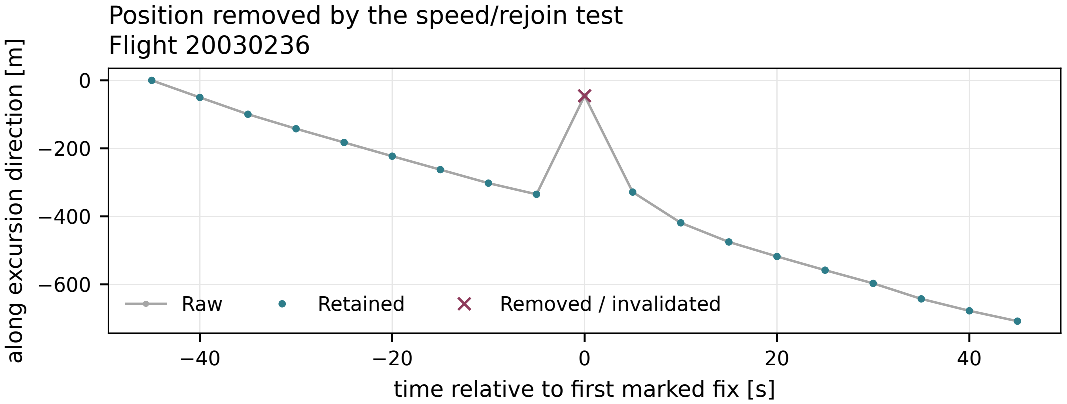
A plausible-looking coordinate can still have the wrong temporal support. Conversely, a sharp turn can disagree with a local median without being a recording defect.

Which structural failures can be detected directly, and which require evidence about the physical trajectory?

Timestamp and coordinate bookkeeping precede geometric decisions

Use the repository distinction between on-the-trend structural defects and off-the-trend geometric defects. The parser decodes a record; cleaning determines admissible ordering and physical reachability. A local median alone cannot establish whether a flight manoeuvre is real.

A Hampel flag describes a departure; the reach/rejoin rule removes it



Which cases can contaminate the median without defeating the reachability check?

Real flight 20030236; the gate uses geodetic distances. Identifier formula and assumptions.

A Hampel flag describes a departure; the reach/rejoin rule removes it

A median-based geometric flag and an absolute reachability decision answer different questions. In the actual example the removed point is far off the local course. But a tight turn can also differ from a local median, so the flag alone must not remove it. A corrupt block longer than the local window can move the median into the corruption; requiring both tests would then miss it. Any impossible step left after the scan becomes a boundary. The initial anchor needs a separate whole-flight reach check because the forward scan does not reconsider it.

The identifier's scale is a convention, not a false-alarm probability

$$r_k = d\big((\varphi_k, \lambda_k), (\widetilde{\varphi}_k, \widetilde{\lambda}_k)\big), \quad \sigma_k = 1.4826\,\mathrm{median}_{|t_j - t_k| \le 20\,\mathrm{s}}\, r_j.$$

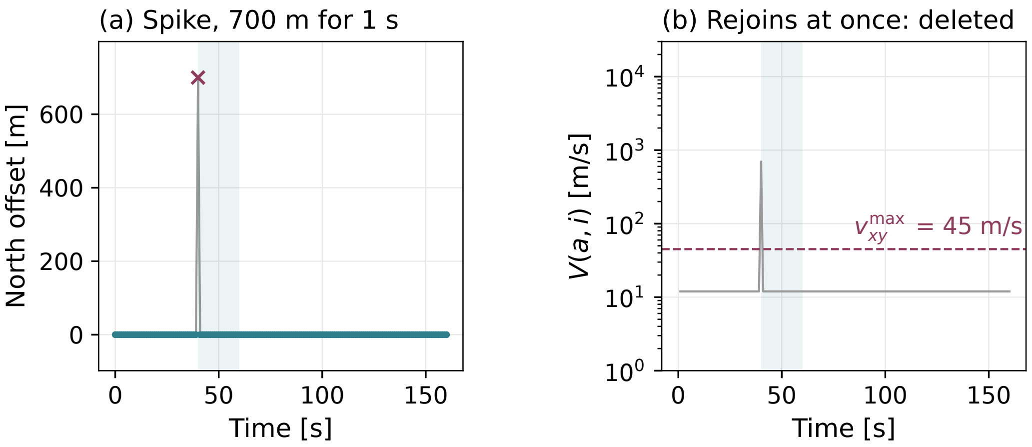
$$\text{Flag when } r_k > \max(5\sigma_k, 15\text{ m}), \quad \text{with at least five fixes in the window.}$$

- Component-wise medians define the local reference position; residuals use horizontal great-circle distance.
- The factor 1.4826 is the one-dimensional Gaussian MAD normalization. For these radial residuals it is an operational scale convention.
- The absolute reach/rejoin gate acts independently of this diagnostic flag.

Identifier terminology: Davies & Gather (1993), Example 2.2. Thresholds and GPS removal rules are choices of this analysis.

The identifier's scale is a convention, not a false-alarm probability

An isolated unreachable fix has an immediate reachable rejoin

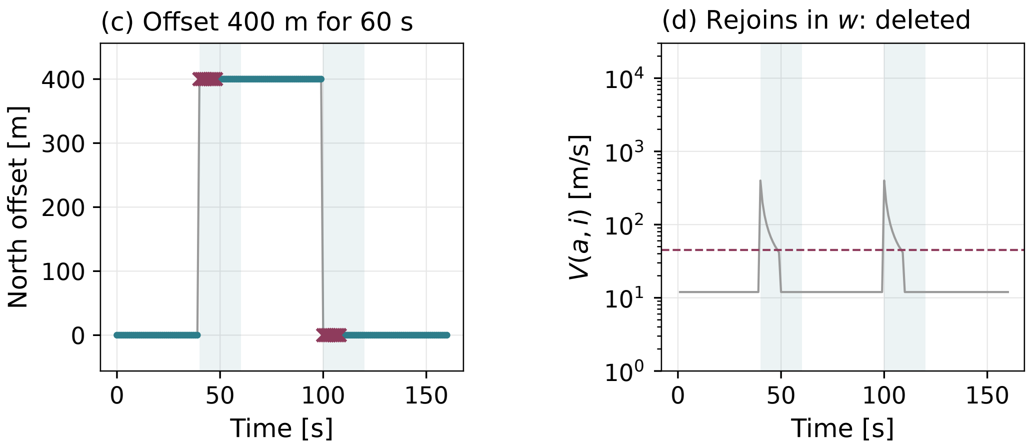


Does deleting the suspect fix restore reachability from the last accepted anchor?

An isolated unreachable fix has an immediate reachable rejoin

The synthetic 700-metre spike lasts one second. Both original panels are retained: position and chord speed from the accepted anchor. The horizontal working bound is 45 metres per second for paragliders, 55 for hang gliders. This example illustrates the implemented rule rather than an independent sensor label.

A moderate offset can become reachable within the block horizon

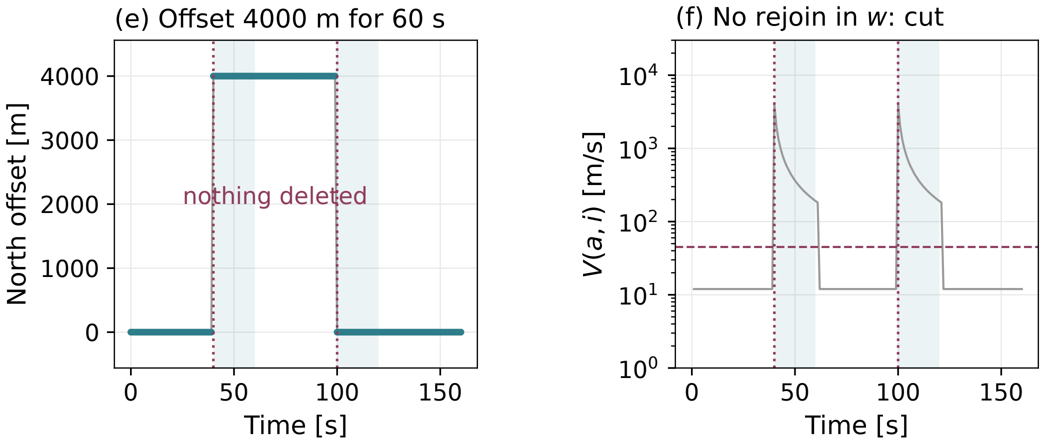


How does keeping the anchor fixed change which early fixes are removed?

A moderate offset can become reachable within the block horizon

The 400-metre offset is reachable within the twenty-second block horizon as allowed displacement grows with elapsed time. The scan removes the shortest block that restores reachability, rather than all fixes that disagree with the original trend. This is a stated limitation of the operational gate.

Without a reachable rejoin, the scan imposes a boundary

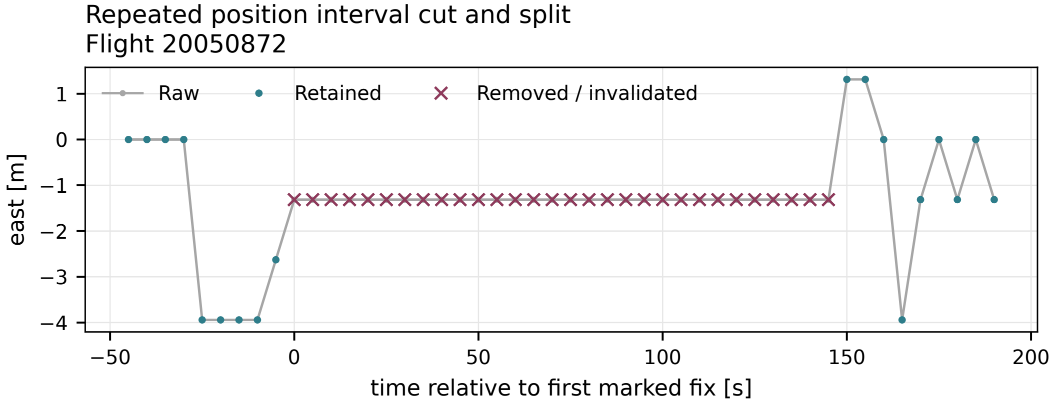


What information remains unresolved after a large reacquisition offset?

Without a reachable rejoin, the scan imposes a boundary

The four-kilometre offset cannot rejoin within the block horizon. The implementation imposes boundaries rather than silently treating the whole interval as an isolated spike. Such a split prevents cross-boundary increments but does not independently validate the new absolute position.

A frozen-position candidate is removed and split



How would genuine slow motion differ from this frozen-position signature?

Real flight 20050872; the removed interval remains a gap. Complete witness branches.

A frozen-position candidate is removed and split

The example stays in a small spatial region while time advances. Extent, duration and witness must agree. An exact coordinate repeat supplies a separate signature. With an eligible barometer, its local coverage must be complete and the first/last quarter medians close. Without an eligible barometer, a majority of invalid GNSS declarations or missing GNSS altitudes is the fallback. With globally eligible but locally incomplete pressure, the barometric branch abstains instead of switching fallback. Quantization, genuine slow motion and shared logger failures are possible confounders.

The complete frozen-position witness

Require $D_I < 15$ m, $T_I \geq 60$ s and $W_I = 1$. Here D_I is the horizontal bounding-box diagonal.

Let E_I mean exact coordinate repetition; d_I is the fraction with invalid GNSS declaration or missing GNSS altitude. Then

$$W_I = E_I \vee \begin{cases} |\Delta b_I| < 5\text{ m}, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1,\ B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0. \end{cases}$$

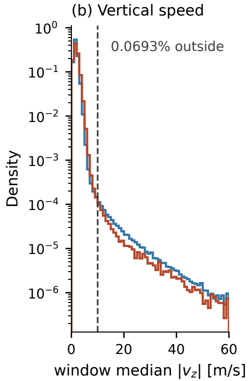
Δb_I is the last-quarter minus first-quarter pressure-altitude median.

For interior ground detection, the pressure test instead uses a detrended 5–95% span and a slope guard, with sustained low speed for at least 10 min. Missing local pressure always causes that guard to abstain.

Small net pressure change can hide an excursion; distinct sensors do not guarantee independent recorder errors. Shorter qualifying slow intervals are flagged and retained.

The complete frozen-position witness

The vertical-speed threshold is 10 m/s



The threshold lies near the slower-decaying tail and the crossing of the two discipline densities around 9–10 m/s.

Above 10 m/s: 0.069% of window medians in the diagnostic sample.

This is an exceedance fraction, not the fraction of altitudes removed. Both neighbouring decisions still enter the rule.

Does changing the threshold materially alter vertical features or phase statistics?

The curve motivates an operational cutoff. It does not establish a physical speed maximum or prove that every larger value is an outlier.

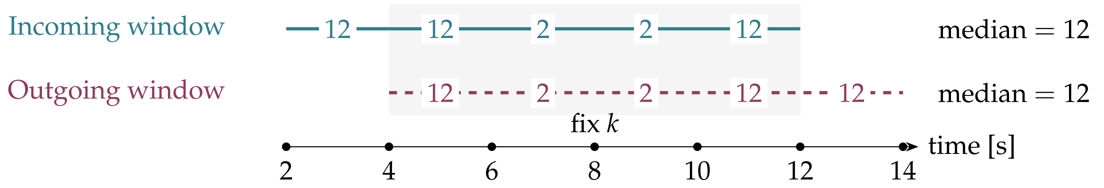
The vertical-speed threshold is 10 m/s

Use the requested graphical criterion without overinterpreting it. A histogram knee or density crossing can guide a working threshold, but its shape depends on heterogeneous physical motion and recorder errors. The median alone cannot distinguish an exceptional descent from a defective altitude interval. The sensitivity of downstream observables matters more than the small marginal exceedance fraction.

Two window medians determine one altitude decision

With at least five finite steps in a window,

$$u_j = \operatorname{median}_{|m_i - m_j| \le 5\,\mathrm{s}} |v^{\mathrm{obs}}_{z,i}|, \quad m_i = (t_i + t_{i+1})/2; \qquad z_k \to \text{missing if } u_{k-1} > 10 \wedge u_k > 10.$$



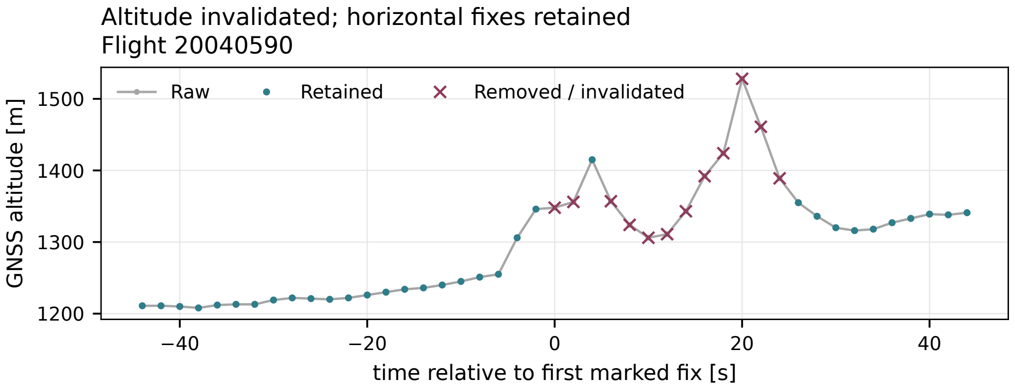
Both medians exceed the threshold although the two steps touching fix k are only 2 m/s. This is a neighbourhood decision.

Schematic with 2-s cadence; speeds in m/s. Below five finite steps, $u_j = |v^{\mathrm{obs}}_{z,j}|$ instead of the median. Equality at 10 m/s does not trigger invalidation.

Two window medians determine one altitude decision

This is the diagram to explain the two formulae without hiding their limitation. Windows are centred on step midpoints, not on fixes. In the example the two central touching steps have magnitudes two, yet the surrounding three large values make each median twelve. Both tests are needed, but their heavily overlapping observations are not independent confirmation. The rule invalidates only altitude, leaving time and horizontal position. The sparse-window fallback uses the step magnitude if fewer than five finite speeds are available. With adequate support, an isolated out-and-back spike can leave the median low and needs the separate opposite-sign step test.

Altitude invalidation precedes the full-trajectory support decision



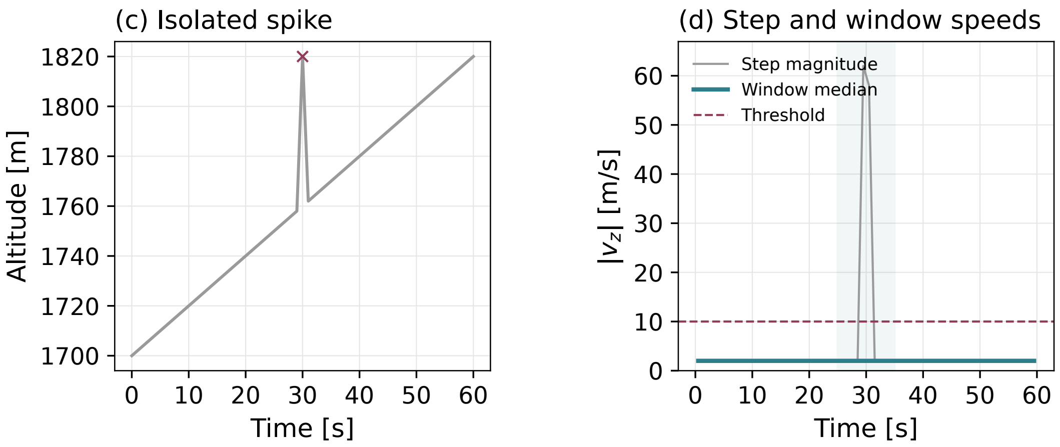
What evidence could independently distinguish an altitude defect from a genuine manoeuvre?

Real paraglider flight 20040590. Synthetic spike and sustained-excess examples isolate what the separate tests do.

Altitude invalidation precedes the full-trajectory support decision

Use the real example to distinguish channel invalidation from removal of the whole fix. An altitude spike is handled by an out-and-back rule with excessive bounding steps of opposite sign. The sustained-excess rule uses the two window statistics just explained. The plotted crosses are the output of the same code used for cleaning. The figure verifies that the implementation and description agree, but we still cannot infer a false-detection rate from these selected examples.

An isolated altitude spike can leave a populated-window median low

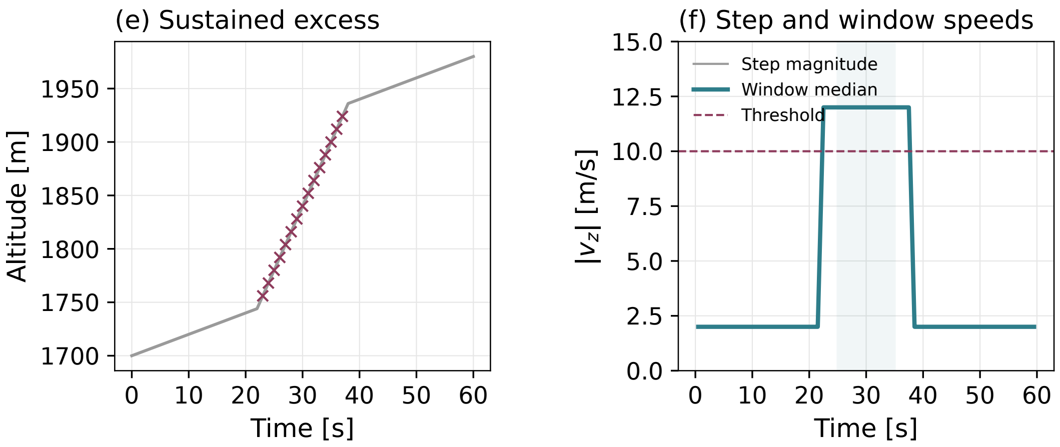


What evidence does the separate opposite-sign bounding-step test add?

An isolated altitude spike can leave a populated-window median low

The synthetic example is processed by the current implementation. Sparse windows fall back to individual step magnitudes. An algorithmic synthetic example isolates a branch of the rule; it does not estimate a detector error rate.

Sustained excess raises both neighbouring window statistics

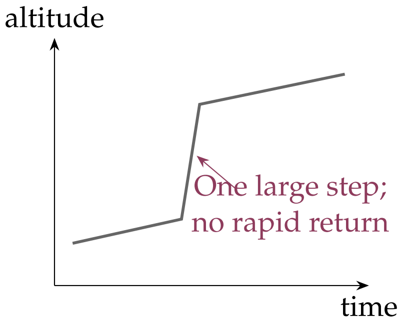


Which plausible physical descents could also satisfy this operational criterion?

Sustained excess raises both neighbouring window statistics

Both neighbouring robust statistics enter the decision, and their windows overlap. They are not independent confirmations. Altitude invalidation is followed by the bounded-support resampling decision.

An unreturned altitude step remains an unresolved case



A single jump may leave the median low and fail the out-and-back test.

Candidate counters occur in 47,440 para / 1,350 hang retained-flight records, before later trimming and segment selection.

A targeted check found candidate times inside usable phase-feature windows in all eight inspected examples.

Compare exclusion or splitting near candidates before deciding whether an automatic correction is justified.

Schematic. Candidate counts do not count confirmed sensor defects surviving in the final trajectories. No automatic level-shift correction is implemented.

An unreturned altitude step remains an unresolved case

This is one of the most consequential open cleaning questions for segmentation. The detector records an excessive step without a return within 30 percent of its amplitude among the next five available fix positions with finite altitude. That short observation horizon does not prove a persistent sensor offset. The counter is computed before later cuts. A slowly varying noise process or real descent can also create such a candidate. An offset can produce a derivative transient after smoothing even when reconstruction masks are false. The eight targeted examples show possibility, not an archive-wide frequency of corrupted features.

Outer trimming uses a sustained horizontal-speed criterion

$$v_{xy} > 5\,\mathrm{m\,s^{-1}} \quad \text{for} \quad 30\,\mathrm{s}.$$

Read forward for onset and backward for the final airborne interval. Observed step speeds supply the evidence; slow periods inside the flight do not set these outer boundaries.

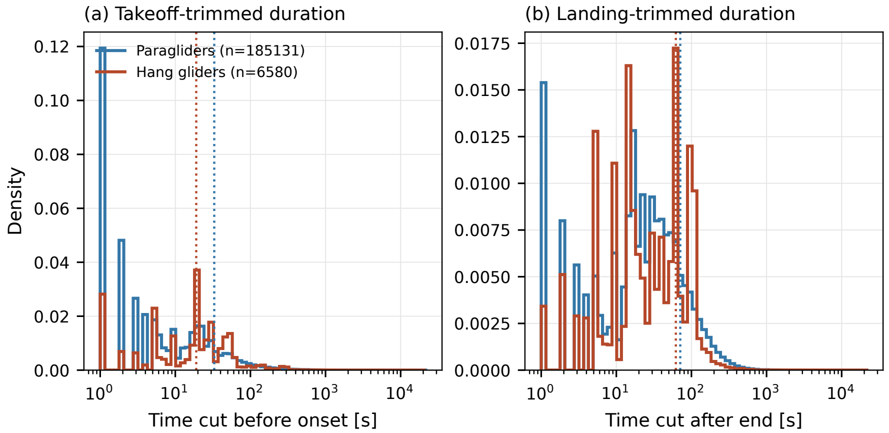
Slow airborne motion can be clipped at an end, while sustained fast ground movement can be retained. The persistence time is not an error bound on take-off or landing.

Which boundary cases should be inspected independently of the algorithm's own decision?

Outer trimming uses a sustained horizontal-speed criterion

There is no claim that the onset is accurate to thirty seconds. A logger may record ground movement before takeoff or a long stationary period after landing. Flight records with no sustained high-speed stretch are rejected at this stage.

Time removed before onset and after landing has a long tail



Do the longest removed intervals reflect continued ground logging or slow flight?

Time removed before onset and after landing has a long tail

These counts cover flights reaching a defined trimming window. They do not describe every catalogue entry or only the final retained population. The raw census was rescanned under the current implementation; the cached and rescanned outputs agreed exactly. A long discarded duration is not automatically a large boundary error.

Interior ground intervals require joint speed and barometric evidence

At least ten minutes of low horizontal speed, complete eligible pressure coverage, a small detrended pressure-altitude span, and a bounded fitted vertical trend.

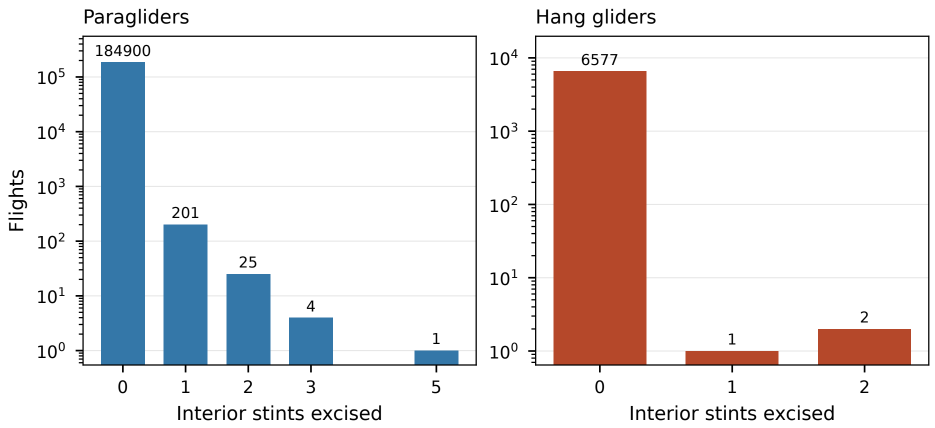
Detrending alone must not erase a genuine steady climb. Without adequate barometry, the detector abstains.

Can ridge or thermal flight meet these conditions, and how would an independent label distinguish it?

Interior ground intervals require joint speed and barometric evidence

The rule removes an interval and imposes a segment boundary, rather than filling it. There are 231 affected paraglider flights and three hang-glider flights in the current trimming census. Shorter qualifying intervals reaching the frozen-duration threshold are flagged but retained; their planned sensitivity analysis has not yet been performed.

Interior excisions affect a small part of the full trimming census



Does a small affected count also imply a small effect on phase statistics?

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Interior excisions affect a small part of the full trimming census

There are 231 affected paraglider and three hang-glider records among those reaching the trimming census. The count is not an independently labelled false-positive rate. Every excision creates a boundary.

Flight-level gates define the study population

Quantity	Working restriction
Airborne duration	at least 40 minutes
Observed path length	at least 20 km
Altitude range	at least 600 m
Elapsed span	at most 16 hours

Further integrity and reach checks apply. Segment acceptance follows separately.

Which valid short flights or low-vertical-excursion flights are excluded from the questions we can answer?

Flight-level gates define the study population

These thresholds are operational population restrictions. They should not be interpreted as universal definitions of soaring or hard physical limits. Following interior trimming, parent-flight duration and path combine the remaining blocks; they are not reapplied as forty-minute/twenty-kilometre gates to every retained segment.

Convert geographic coordinates before measuring horizontal motion

$$(\varphi, \lambda, h) \longrightarrow \mathbf{r}_{\mathrm{ECEF}} \longrightarrow \begin{pmatrix} E \\ N \\ U \end{pmatrix} = R(\varphi_0, \lambda_0)[\mathbf{r}_{\mathrm{ECEF}} - \mathbf{r}_{\mathrm{ECEF},0}].$$

The local orthonormal basis is fixed at the flight origin. East and north are measured in metres, avoiding subtraction of latitude and longitude as if they were Cartesian distances.

On which route lengths would a fixed tangent frame materially affect a geographical comparison?

Convert geographic coordinates before measuring horizontal motion

Separate the coordinate transformation from the altitude channel used for vertical analysis. The geodesy appendix derives the ECEF/ENU rotation and its geometry. A fixed local frame does not follow the changing tangent plane along a very long route.

The local basis uses the origin's geodetic coordinates

$$R(\varphi_0, \lambda_0) = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix}.$$

Subtract the origin in ECEF, then rotate. Geodetic latitude follows the ellipsoid normal; geocentric latitude follows a radius from the Earth's centre.

The IGC latitude already supplies the geodetic angle used in this transformation.

A coordinate convention can create an apparent displacement error before any physical model is fitted.

The local basis uses the origin's geodetic coordinates

This is a fixed orthonormal rotation of the three-dimensional ECEF displacement, not an independently refitted frame at each lag. On the ellipsoid, geodetic and geocentric latitude differ. The complete derivation and approximations belong to the thesis geodesy appendix and can be discussed without treating the Earth as locally flat over arbitrary distances.

Rotated Up and the recorded altitude are different coordinates

$$(E, N, U) = R\,[\mathbf{r}_{\mathrm{ECEF}} - \mathbf{r}_{\mathrm{ECEF},0}], \qquad z = h_{\mathrm{GNSS,logged}}.$$

The stored vertical channel is logged altitude, not height above the origin's fixed tangent plane. It is not reset to zero at launch.

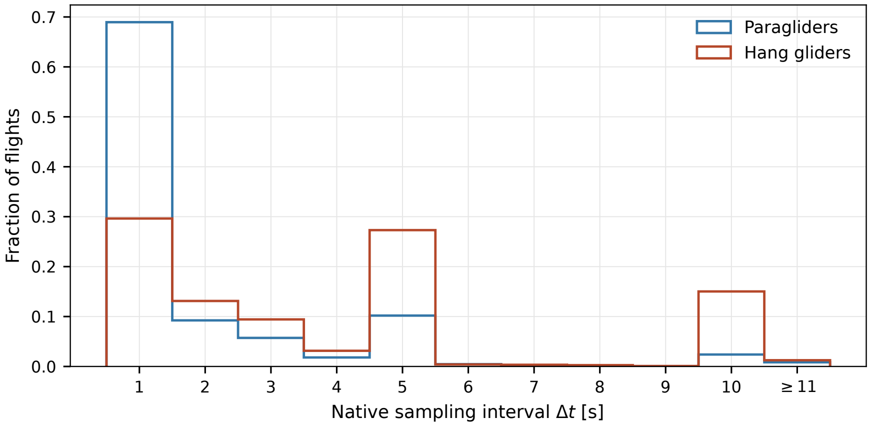
GNSS altitude supplies a height proxy for the ECEF conversion. Temporary height interpolation used by that conversion does not validate a missing vertical interval; the later gap rule still controls which complete trajectory segments survive.

Which effects depend on an absolute altitude datum, and which cancel in within-segment increments?

Rotated Up and the recorded altitude are different coordinates

Keeping this distinction avoids interpreting Earth-curvature sagitta as a physical descent. A constant vertical bias cancels from vertical differences; spatial or temporal changes do not. The implementation preserves missing-altitude support for subsequent resampling decisions even when a height proxy was needed to calculate horizontal coordinates.

Recorder cadence varies across the raw archive



Which native cadences resolve the shortest scales used later?

Recorder cadence varies across the raw archive

Preprocessing retains the native median cadence, while Chapter 3 applies a separate common ten-second grid to eligible segments. Horizontal interpolation is shape-preserving cubic and altitude interpolation linear, within the bounded support. A finer numerical grid would not restore missing physical resolution.

The same duration bound governs horizontal and vertical support

$$g_{\max} = \min(10\Delta t,\ \max(20\,\mathrm{s}, 2\Delta t))\,.$$

Native cadence Δt	1 s	5 s	15 s
Gap bound $g_{\max}$	10 s	20 s	30 s

At or below the bound: interpolate and flag. Above it: split the complete trajectory. For altitude, compare consecutive finite source readings, not the length of a flag run.

How do threshold changes affect both reconstructed motion and which long-lag windows survive?

The same duration bound governs horizontal and vertical support

The cap is not universally twenty seconds. Equality at the actual cadence-dependent bound is admitted. A long altitude-only hole removes the intermediate horizontal fixes as well; missing prefixes and suffixes are excluded without extrapolation. Mandatory cleaning boundaries always remain boundaries.

Long altitude holes split the full trajectory

Flight 20171597 has no admissible vertical support between 6805 s and 7450 s. The 645-s source bracket exceeds the gap bound: one supported segment ends at 6805 s and the next starts at 7450 s.

No fixes from the unsupported interior enter the retained trajectory. Its exclusion is recorded, and the supported pieces undergo the usual segment gates.

Is preserving a complete three-coordinate segment the appropriate trade-off when horizontal position alone remains available?

Raw-flight reprocessing witness for cleaning 2.3.0; the longest remaining altitude-reconstruction flag run in this flight is 1 s.

Long altitude holes split the full trajectory

Explain the loss of horizontal coverage explicitly. The benefit is a consistent contract for subsequent three-dimensional observables; the cost is censoring and a reduction in usable temporal support. This witness verifies one concrete behavior, while the full structural and reproducibility checks cover different aspects of the pipeline.

A segment boundary preserves the flight clock and coordinate origin

Supported pieces keep their parent time and origin. A new segment prevents an increment stencil from joining observations across an unsupported interval.

The observed path length sums within segments. The net displacement between the first and last retained endpoints can still span an unobserved interval.

Which observables depend only on within-segment motion, and which still depend on the separation between recorded pieces?

A segment boundary preserves the flight clock and coordinate origin

This distinction matters for the launch-referenced second moments and the observed-path closure descriptor. Dividing endpoint separation by only the observed path length need not produce a number bounded by one. Splitting removes unsupported interpolation; it does not independently verify every reacquired absolute position.

Segment acceptance requires duration and complete local support

Requirement	Current rule
Segment duration	at least 90 s
Time-interpolated grid fraction	at most 10%
Finite altitude support	each source bracket within $g_{\max}$
Local polynomial	enough samples for its configured window

One parent flight can retain several segments or lose them all. Its original 40-minute and 20-km gates are not reapplied to every segment.

How does the retained population change when the minimum duration or interpolation fraction changes?

Segment acceptance requires duration and complete local support

The ten-percent criterion is the time-grid interpolation fraction; it is not an extra ten-percent cap on altitude-only reconstructed values. The latter obey the finite-source gap bound. Report segment counts separately from parent-flight counts, and distinguish conditional loss at this stage from the overall rejection rate.

Quality flags describe support; every retained coordinate is smoothed

Flag or boundary	Interpretation
Interpolated	Reconstructed sampling support
Altitude reconstructed	Short missing-altitude support; long holes are split
Edge	Boundary treatment of the local polynomial window
Segment boundary	Increment and feature windows must not cross it

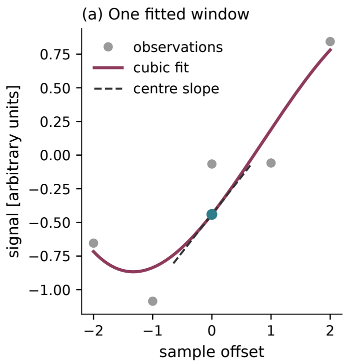
An unflagged interior fix is still part of the filtered trajectory. A sequence of nearest-fix flags is not a direct measurement of finite-source altitude spacing.

Which masks are needed by each observable, and which plausible defects can remain unflagged?

Quality flags describe support; every retained coordinate is smoothed

The independent irregular-grid audit found four flag runs longer than twenty seconds even though every source-altitude bracket obeyed the gap bound. That was a distinction between flag propagation and source support, not evidence of a long interpolated altitude hole. Level-shift candidates remain a separate uncertainty.

One local polynomial estimates position and derivatives



For offsets $u = -m, \ldots, m$,

$$P_i(u) = \sum_{k=0}^{3} a_k(i) u^k,$$

$$\widehat{x}_i = a_0,$$
$$\widehat{v}_i = a_1 / \Delta t,$$
$$\widehat{a}_i = 2a_2 / \Delta t^2.$$

The same fit is applied separately to E, N, z inside each segment.

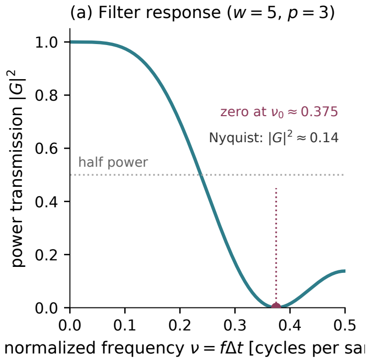
Smoothing controls derivative noise, but it also changes resolved motion and propagates reconstructed values through its windows.

Synthetic illustration of the implemented Savitzky–Golay fit. Savitzky & Golay (1964).

One local polynomial estimates position and derivatives

The polynomial is fitted in sample offsets, so physical derivatives require division by cadence and its square. A cubic is the present modelling choice; a quadratic would already permit nonzero acceleration. The coefficients are computed using the numerical library rather than explicitly inverting a normal-equation matrix. The example is synthetic and only illustrates the estimator. Real GNSS noise can be correlated and nonstationary, and a plausible filtered curve is not proof that a manoeuvre has been preserved.

The nominal 5-s timescale does not mean a 5-s window for every logger



$$w = \max\{\lceil 5\,\mathrm{s}/\Delta t\rceil_{\mathrm{odd}}, 5\}.$$

Five samples span:

Cadence	Window span
1 s	4 s
3 s	12 s
5 s	20 s

The filter's frequency response depends on cadence.

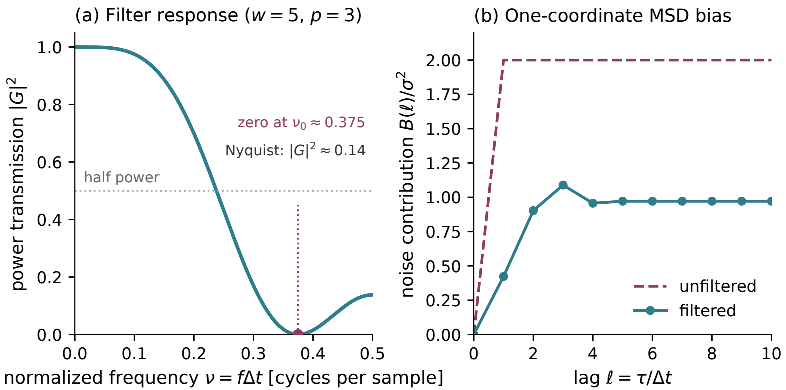
Quantify the sensitivity of 10-s transport and phase features to cadence and the smoothing window.

The observed spectral knee motivates a working timescale; it is not the exact cutoff of the Savitzky–Golay filter. The response shown is the interior value-filter response.

The nominal 5-s timescale does not mean a 5-s window for every logger

Even though the configured timescale is five seconds, the minimum five-sample window can span twenty seconds for a five-second logger. This makes the 10-second endpoint of Chapter 3 potentially sensitive to preprocessing. The current horizontal and vertical timescales are separate parameters that happen to have equal values. The exact filter response was computed from the adopted coefficients. What is still missing is a targeted comparison of the scientific observables under alternative cadences and windows, not another statement that the filter is smooth.

The value filter changes the contribution of position noise



Left: the interior value-filter response. Right: its MSD contribution for independent white position noise of variance σ^2, in one coordinate.

How would correlated recording errors change this illustrative noise calculation?

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The value filter changes the contribution of position noise

The right panel is a stated white-noise model, not a measured decomposition of the flight MSD. Neither panel plots the derivative transfer function. Position and derivatives come from the same local polynomial but have different coefficients. A nominal five-second timescale gives a five-sample minimum window, whose physical span depends on cadence.

Smoothing modifies the measured displacement variance

$$\widehat{x}_j = \sum_k c_k x_{j+k}, \qquad \Delta_\ell \widehat{x}_j = \sum_k c_k \Delta_\ell x_{j+k}.$$

Consequently,

$$\mathbb{E}[(\Delta_\ell \widehat{x})^2] = \sum_{k,m} c_k c_m \, \mathbb{E}[\Delta_\ell x_{j+k} \Delta_\ell x_{j+m}].$$

The effect depends on correlations as well as the filter coefficients. A visually smoother trajectory does not establish that its short-lag MSD is unchanged.

On a common set of flights, which fitted ranges remain stable under cadence and window changes?

Smoothing modifies the measured displacement variance

This identity is for an interior linear convolution; boundary fitting needs separate treatment. It does not assume independent increments or declare the unfiltered observation to be the true path. The sensitivity comparison must separate changes to measured motion from changes to the set of admitted flights.

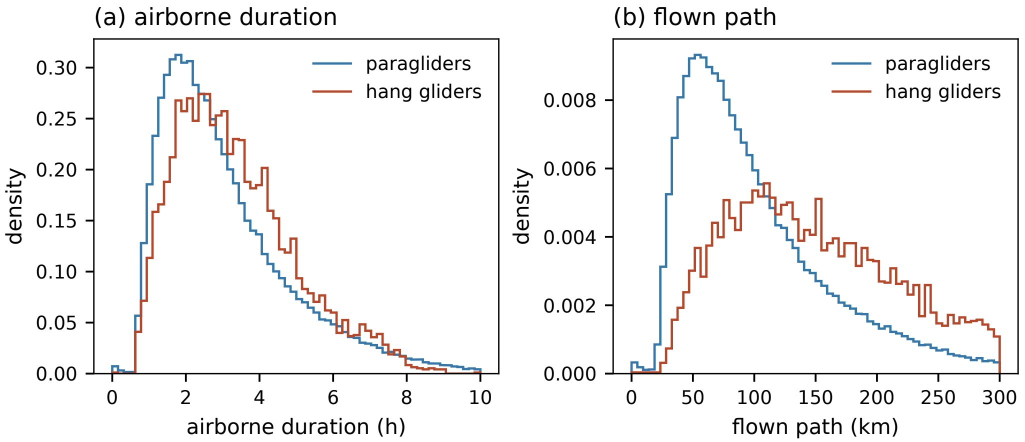
The complete first-failure census

Why the flight was dropped	paragliders	hang gliders
unreadable track	27	0
no usable GNSS altitude	697	135
never airborne	197	1
integrity gate	550	53
shorter than the duration cut	2,084	79
longer than the duration cut	3	0
path below the floor	173	1
altitude activity below the floor	4,899	155
out of reach of its own first fix	217	5
no segment survived resampling	20,798	193
no segment carries the smoothing window	1	0
retained	**156,406** (84.1%)	**6,094** (90.7%)

Current clean archive, pipeline version 2.3.0. All numerical entries are imported from the frozen generated table.

The complete first-failure census

Elapsed record span and observed path use different gap conventions

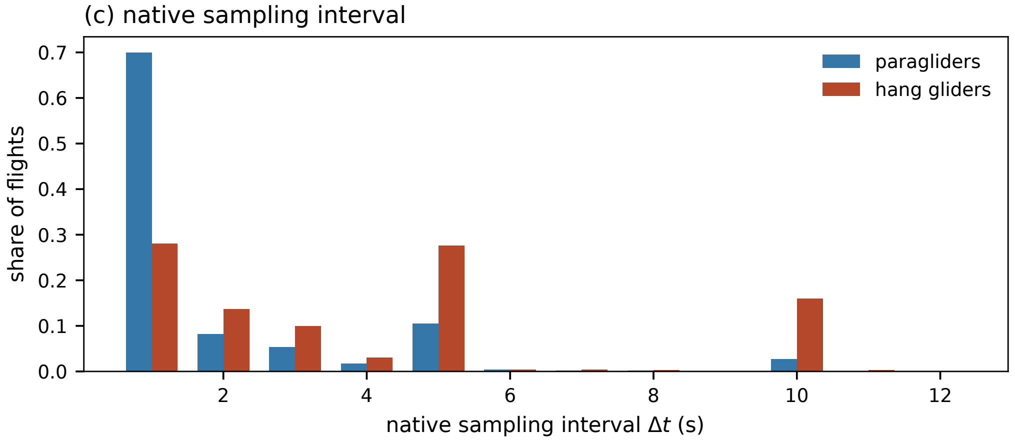


Duration spans the first to last retained fix, including gaps; path sums only within segments.

Elapsed record span and observed path use different gap conventions

The plotted duration is the elapsed span from first to last retained fix, including gaps. The path sums within retained segments and excludes missing intervals. Chapter 3 duration cohorts instead sum the retained segment spans. Later segment losses can reduce duration and path below the earlier parent-flight gates. The histograms preserve the original bins and normalization.

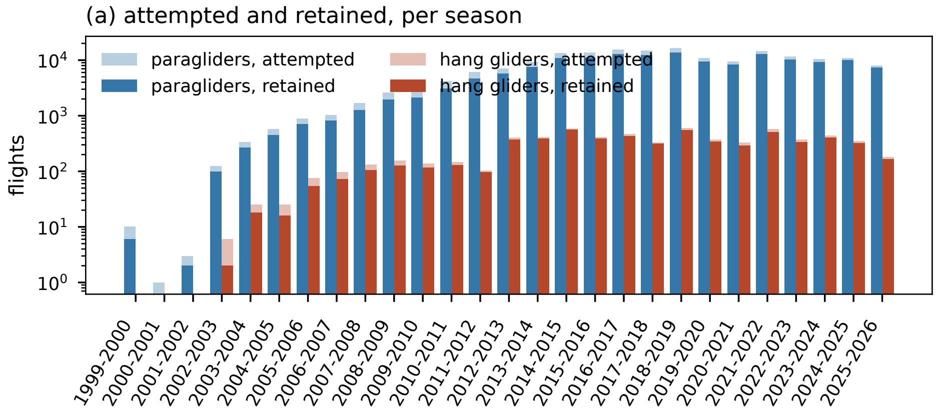
Retained cadences define the available temporal resolution



Which results could be sensitive to the changing mixture of recorder intervals?

Retained cadences define the available temporal resolution

The plotted cadence distribution concerns retained flights, unlike the broader raw-record diagnostic shown earlier. Higher counts at a cadence do not prove equal device quality or equal coverage of sites and seasons.

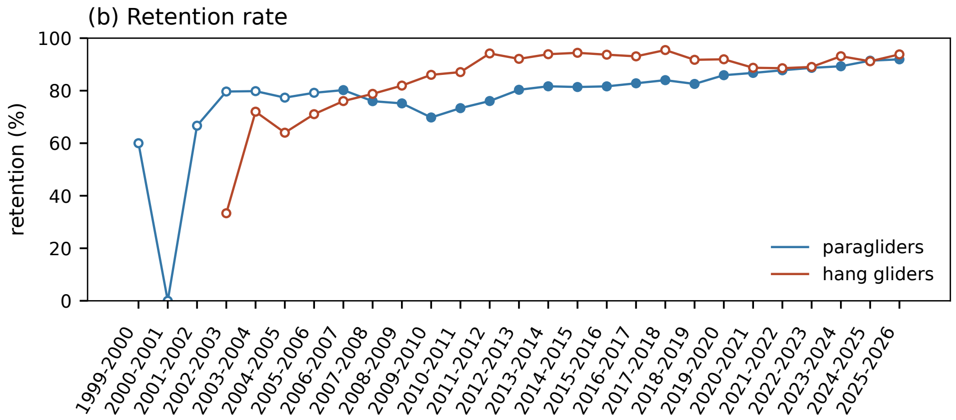


Which temporal comparisons also change equipment, geography or logging technology?

Participation and available tracks change across seasons

Attempted and retained counts are on the original logarithmic scale. The lower original season labels are appended without changing horizontal coordinates.

Similar aggregate retention can conceal different selection

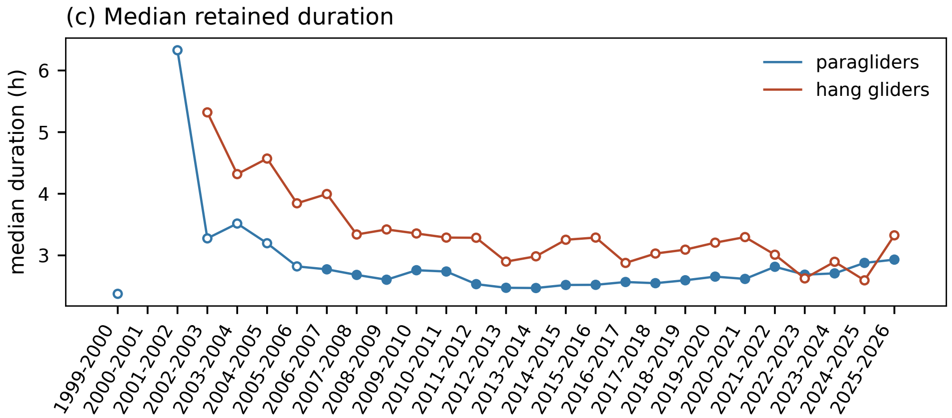


Are cadence and regional retention stable within the larger seasons?

Similar aggregate retention can conceal different selection

Open markers indicate seasons with fewer than one thousand flights. Similar overall retention is not evidence that every subgroup is selected at the same rate.

The duration of retained records also changes with season



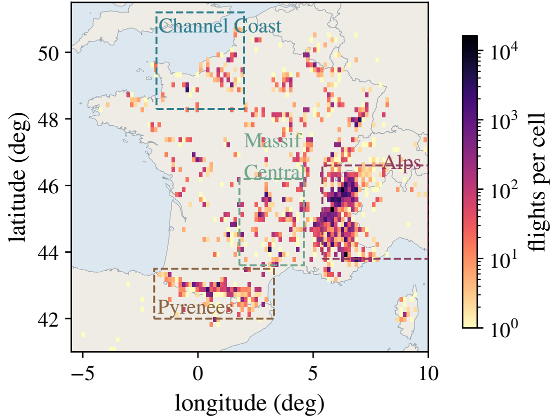
Would matching season change the apparent equipment or regional contrast?

The duration of retained records also changes with season

These medians describe survivors. They combine real flight duration, who declares and records flights, device support and cleaning selection.

Geographic coverage is part of the sampling problem

(a) France — 94.8% of flights



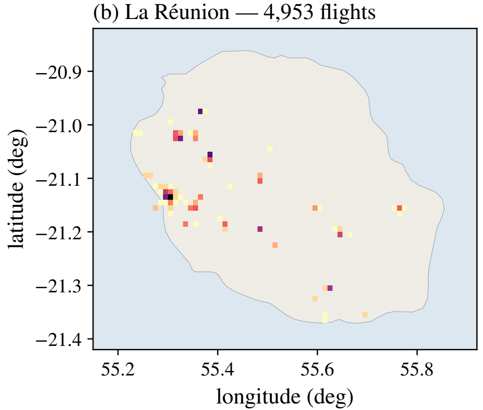
Launch sites and flight characteristics are unevenly represented.

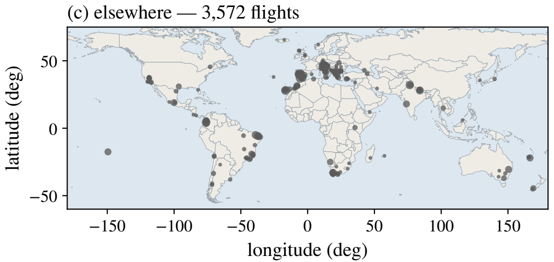
Equipment, season, recording technology and environment can change together.

Directional transport is analysed in Chapter 3; this chapter establishes which observations reach that analysis.

Geographic coverage is part of the sampling problem

La Réunion and other sites broaden the observed population



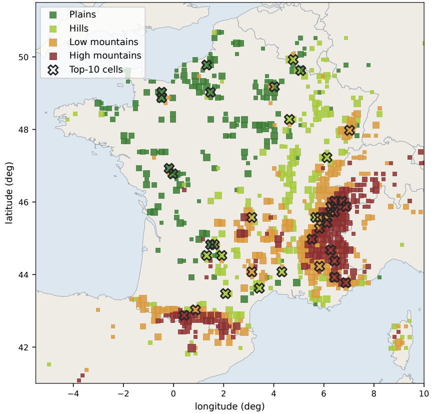


Which pooled results combine different climates, seasons and flight environments?

La Réunion and other sites broaden the observed population

Both disciplines are pooled. The La Reunion density uses the original 0.01-degree cells; the elsewhere map uses 0.5-degree site cells with marker area reflecting their count. Bins, geographic extents and plotted values are unchanged from the chapter; only typography and layout have been adapted.

Logged starting altitude does not measure local terrain relief



The map groups first raw altitudes: GNSS where available, with pressure fallback for this exploratory map only.

Its labels (plains, hills, low and high mountains) refer to altitude bands. They do not measure slope, relief or exposure along a route.

Would a terrain model and launch-location check change the environmental strata?

Logged starting altitude does not measure local terrain relief

The first raw fix can precede take-off, occur after the flight has started, or contain sensor settling. Pressure fallback does not enter the cleaned vertical trajectory. A low starting altitude does not rule out ridge or cliff soaring, and a high plateau can have low local relief.

What has been checked, and what still needs empirical validation?

Completed

- Cleaning 2.3.0 and identified numerical products.
- Rule tests, reconstruction masks and trajectory examples.
- Complete structural scan: 1.404 billion retained fixes.
- Reconciliation of numerical reports and manuscript definitions.

Still open

- Independent detector error rates.
- Observable sensitivity to thresholds, smoothing and cadence.
- Candidate altitude steps and retained suspect intervals.
- GNSS height-datum consistency where it affects conclusions.

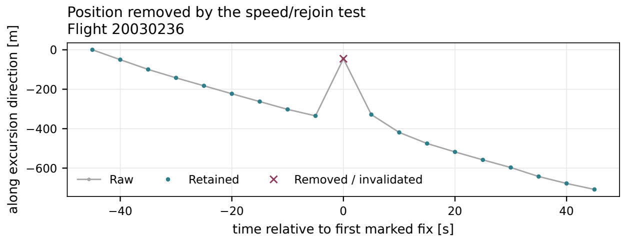
Prioritize checks that could change the transport conclusions or the phase features.

Numerical reproducibility and mathematical consistency are necessary. They do not establish physical correctness of every retained fix or every selection rule.

What has been checked, and what still needs empirical validation?

The companion manifest identifies the completed numerical run, its source hashes and the reviewed manuscript. The full structural scan checks operational tolerances; it reports small residual anomalies and is not an external sensor reference. Retain the separate reproduction and targeted-rule audits in the evidence manifest. Independent labelled examples and sensitivity of scientific observables remain open.

Inspect one of the records together



Open the Python viewer

Paraglider flight **20030236**
Position-spike example around 4410 s after the raw record begins.

Compare the raw and current processed trajectory, then zoom into the suspect interval.

Local macOS link; the companion launcher must be registered. The viewer can open without the SSD; this raw flight needs the connected archive. Offline fallback: the exact frozen example above.

Inspect one of the records together

Use the viewer if raw data and the registered local launcher are available. The frozen example is also in the slide for offline discussion. Loading one flight may recompute that flight with the current code; it does not rebuild the archive. Independently assess the neighbouring trajectory before using an algorithmic decision as a label.

Which cleaning uncertainties should we resolve first?

1. **Vertical reliability.**
   Compare observables with and without neighbourhoods of candidate altitude steps; assess sensitivity to the adopted vertical gap bound.
2. **Sensitivity of the scientific results.**
   Vary the 10-m/s threshold and smoothing/cadence choices on a fixed comparison population; report changes in transport and phase features.
3. **Independent evidence for difficult decisions.**
   Review paired-channel examples and realistic injected defects; distinguish acceptance criteria from measured error rates.

The trajectories are reproducible. Their remaining uncertainties must be quantified where they affect the physics.

Which cleaning uncertainties should we resolve first?

The vertical gap cap has been implemented. The remaining question is its scientific sensitivity, alongside altitude-step candidates, cadence and smoothing. Compare common populations where possible and report changes in admission separately. Agree which independently labelled or injected defects would most improve confidence in the phase features.

Transport plots must retain explicit support and populations

- Retained trajectories: native uniform segments, fixed flight origin and explicit support flags.
- Transport increments: all eligible segments on the stated common grid, with no cross-gap stencil.
- Long-lag comparisons: show changing support and add fixed-population controls.
- Regional interpretations: distinguish observation quality, selection and physical mechanisms.

Which sensitivity analysis would most change confidence in the transport conclusions?

Transport plots must retain explicit support and populations

The cleaning pipeline provides traceable decisions and a coherent observation contract. Structural checks and reproducibility establish consistency with that implementation; independent labels and perturbation analyses address whether the working decisions best preserve the intended physical quantities.

Methods and provenance

Savitzky & Golay (1964)	Polynomial smoothing and derivatives
Fritsch & Butland (1984)	Monotone piecewise cubic interpolation
Davies & Gather (1993)	Hampel identifier

Numerical sources: current Chapter 2 census, paired-channel diagnostic and real cleaning examples. Figure and value snapshots are identified by SHA-256 hashes in the companion manifest.

The cited methods do not establish the validity of the chosen numerical thresholds, the recorder status of individual fixes or the accuracy of downstream phase labels.

Methods and provenance